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Laboratory Investigations

An Ultrastructural Study of Calcium Phosphate Formation in Multilamellar Liposome Suspensions

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Summary. Calcium phosphate precipitation can be induced within liposomes containing buffered inorganic phosphate by the ionophore-mediated loading of calcium ions. Negative staining, positive staining for thin sectioning, and freeze-fracture electron microscopy were used to characterize these synthetic vesicles and to evaluate the liposome-mineral interactions resulting from apatite formation. Suspensions of phosphate (0–50 mM KH_2PO_4)-encapsulated liposomes were prepared from mixtures of phosphatidylcholine, dicetyl phosphate, and cholesterol in the molar ratios of 7:2:1. Precipitation reactions were initiated by first suspending the liposomes in a buffered solution containing calcium (1.3–2.2 mM $Ca(NO_3)_2$) and then adding the cationic ionophore X-537A. All experiments were carried out at 22°C, pH 7.4, and 240 mosm. Transmission electron microscopical analysis showed that the liposome preparation consisted of multilamellar, multicompartmental vesicular structures. The liposomes were typically heterogeneous with respect to both the size and number of phospholipid bilayers surrounding the aqueous cores. In Ca-loaded liposomes, discrete clusters of apatite mineral were present within the lumen, and in close proximity to the inner lipid membranes. These nascent crystallites eventually penetrated the lipid envelope to provide a focus for external precipitation events. Crystalline apatite phases were not observed when the incubation conditions prevented intraliposomal precipitation. The *de novo* calcification of these liposomes had many features in common with the sequence of mineral deposition occurring in matrix vesicle-mediated calcification. These results reinforce the conclusions of earlier chemical and kinetic studies and further support the use of this system as an experimental model for examining the membrane-mineral interactions associated with tissue mineralization.

Key words: TEM — Liposomes — Apatite — Matrix vesicles.

Electron microscopic observations on the initial calcification of epiphyseal cartilage [1, 2], and a number of other mineralizing tissues [3], have indicated that the first crystals of apatite mineral are associated with extracellular matrix vesicles. Although complementary biochemical studies have indicated that vesicle phosphatase activities [4–6] and calcium-binding phospholipids [7, 8] may be linked to this mineralization process, the precise mechanism by which the vesicles focally concentrate calcium and phosphate and induce the subsequent formation of mineral deposits is unresolved.

Recently, liposomes have been used as a model system for *in vitro* analyses of the physiochemical processes associated with matrix vesicle-mediated calcification [9–12]. These studies were an extension of earlier work with liposomes aimed at elucidating the molecular mechanisms underlying dystrophic mineralization [13] where cell injury has been linked to the formation of inorganic solids within membrane-bound compartments [13–16]. Depending upon the experimental conditions employed, the initial mineralization of the matrix vesicle model system was found to occur in either of two locales: the external solution phase [12] or in-

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side the liposomes [9, 10]. External precipitation resulted when dioleoylphosphatidate liposomes were suspended in the appropriate metastable calcium phosphate solutions [12]. Extraliposomal calcification could be bypassed initially in favor of precipitation within the aqueous interior by prefilling phospholipid vesicles (prepared from 7:2:1: molar mixtures of phosphatidylcholine (PC), dicetyl phosphate (DCP), and cholesterol (chol)) with concentrated solutions of soluble phosphate salts and then making the lipid membranes permeable to ionophore-controlled Ca^{2+} fluxes [9]. In time, this nascent mineral acted as the nidus for seeding the external solution phase [10].

As an extension of this latter work, transmission electron microscopy was used in the present study to characterize the 7PC:2DCP:1chol liposome preparation, and to examine the nature of the liposome-mineral relationships resulting from apatite deposition. Negative staining, thin sectioning, and freeze-fracture techniques all revealed closed membranous structures with the classical morphology of multilamellar liposomes. Apatite detected within the liposomes was localized in close juxtaposition to the lipid bilayers. Occasionally, these crystals ruptured the limiting walls of the liposomes, penetrated into the external solution phase, and facilitated extraliposomal precipitation.

Materials and Methods

Sources

Egg yolk phosphatidylcholine, dicetyl phosphate, and cholesterol were obtained from Avanti Polar Lipids, Inc. (Birmingham, AL). Sephadex G-50C gel was supplied by Pharmacia Fine Chemicals (Piscataway, NJ). The ionophore X-537A (lasalocid acid, sodium salt) was purchased from Aldrich Chemical Co. (Milwaukee, WI) and the buffers PIPES (piperazine-N,N'-bis[2-ethane-sulfonic acid]) and HEPES (N-2-hydroxyethylpiperazine-lN'-2-ethanesulfonic acid) were acquired from Sigma Chemical Co. (St. Louis, MO). Tannic acid was obtained from Mallinkcrodt (St. Louis, MO). All other materials used in the preparation of samples for electron microscopy were supplied by Electron Microscopy Sciences (Fort Washington, PA). The above reagents had a stated purity of 99% or better and were used without further purification. All aqueous solutions were prepared in deionized-distilled, CO_2-free water.

Liposome Preparation

Liposomes were prepared as previously described [9, 10] from phosphatidylcholine, dicetyl phosphate, and cholesterol in the molar ratios of 7:2:1. Dry films evaporated under vacuum from chloroform solutions (15 μmol lipid/ml) were dispersed in a PIPES-buffered (50 mM) KCl solution containing 0–50 mM KH_2PO_4 by mechanical agitation until all the lipid was suspended. After a suitable aging period (20 hours at 22°C) unencapsulated solution was removed by gel filtration (Sephadex G-50C) using a HEPES-buffered (10 mM) NaCl solution as the elutant. In some cases the liposome preparations were subjected to rapid, repeated extrusions through multiple capillary channels (Liposor extruder, Lidex Corp., Bedford, MA) before applying them to the column. The eluted fractions richest in liposomes (as measured by optical density at 450 nm, [9]) were collected for the precipitation experiments.

Calcium-liposome interactions were initiated by mixing the pooled liposome fractions with an equivalent volume of a HEPES-buffered (50 mM) solution containing 2.66 or 4.50 mM $Ca(NO_3)_2$ and injecting the carboxylic ionophore X-537A (15 μl/ml of a 0.6 mM ethanolic solution of lasalocid acid, sodium salt) into this mixture. In some experiments, the external solution phase was rendered supersaturated with the addition of inorganic phosphate (1.0–1.5 mM KH_2PO_4) as described previously [10]. All experiments were carried at 22°C, pH 7.4, and 240 mosm.

Preparation of Samples for Electron Microscopy

Samples were processed for electron microscopical examination 5 hours after the initiation of ionophore-mediated calcium uptake. X-ray diffraction and kinetic studies have determined that at this time, amorphous calcium phosphate phases are no longer present and that apatite formation has occurred [9].

Negative Staining. Aliquots of liposome suspensions were placed on copper grids coated with formvar and reinforced with carbon (rendered hydrophilic by ionization under an ultraviolet source). After 1 min, excess fluid was drawn off from the side with filter paper. A drop of staining solution (2% potassium phosphotungstate, pH 7.5, or 2% ammonium molybdate, pH 7.5) was applied to the grid for 30s. Excess stain was removed with absorbent paper and the grids were allowed to dry in air for 30 min.

Positive Staining for Thin Sectioning. Samples (1 ml) of liposome suspensions were fixed with an equal volume of 1% osmium tetroxide in 0.1 M phosphate buffer, pH 7.5, for 48 hours at 4°C. The fixed material was then pre-embedded in an equal volume of 4% agar [17] before removal of any excess fixative. Upon hardening, the liposome-agar pellet was cut into small cubes which were rinsed in phosphate buffer, dehydrated in a graded ethanol series (70, 90, and 100%) and embedded in Spurr's low viscosity resin [18]. In some experiments, 1 ml of lipid suspension was pretreated for 30 min at room temperature with an equal volume of a phosphate-buffered (0.1 M, pH 7.5) solution of tannic acid (0.5%). Tannic acid reacts with the choline component of PC to form a complex which can be stabilized by fixation with osmium tetroxide [19]. After pre-embedding in agar, as described above, the samples were rinsed in three to four changes of phosphate buffer and fixed with OsO_4 (1%) in phosphate buffer (0.1 M, pH 7.5) for 1 hour. Following two washes in buffer and dehydration in a graded ethanol series, the samples were embedded in Spurr's resin. Ultrathin sections (silver-gold) were cut with a diamond knife onto buffered flotation fluid (0. 1 M sodium cacodylate, pH 7.4) [20] and collected on 300-mesh copper grids. The sections were stained briefly with lead citrate [21] or examined unstained.

Freeze Fracture. A small quantity of the liposome suspension was trapped between two freeze-fracture planchettes (gold discs, 3 mm) (Balzers Union, Hudson, NH) and rapidly frozen in liquid freon cooled by liquid nitrogen. The frozen samples were stored in liquid nitrogen until transfer into a freeze etch unit (model 301; Balzers). The specimens were fractured under vacuum (10^{-6} Torr) at -120°C and replicated immediately with a 2–3 nm layer of platinum-carbon at an angle of 45° with respect to the fracture surface. Carbon was then evaporated (10–20 nm) at normal incidence onto the replica for strength. Following cleaning with methanol-chloroform, the replicas were rinsed several times in sterile, distilled water and collected on uncoated, 400 mesh grids.

All specimens were examined in a JEOL 100CX transmission electron microscope operating at 80 kV with the liquid nitrogen-cooled anticontamination device attached.

Results

Both negative-stained (Figs. 1, 2) and osmium-fixed (Figs. 3, 4) preparations of 7PC:2DCP:1chol lipid suspensions revealed a heterogeneous population of closed multilamellar liposomes. Freeze-fractured liposomes revealed a similar morphology (Figs. 5, 6, 7). Cross-fractured replicas of individual liposomes showing numerous concentric, uniform bilayers confirmed that the lamellar configuration was native to the lipid mixture in aqueous suspension (Fig. 7). Freeze fracturing also demonstrated the multicompartmental structure of the phospholipid vesicles with some liposomes appearing as a system of interlocking spheres (Fig. 6).

The average number of lamellae per liposome was 6 with a population spread ranging from 2–30 recorded. Analysis of phospholipid vesicle diameters revealed a mean value of 1.2 μm; liposomes between 0.5 and 2.92 μm accounted for 75% of the total volume. When subjected to the capillary extrusion process, the liposomes exhibited fewer lamellae than their untreated counterparts. The diameters of these lipid vesicles (as determined from micrographs of negatively stained samples) were also reduced (50–800 nm after 9 min recycling through the apparatus). Aside from these differences, these oligolamellar liposomes were comparable in morphology to the untreated preparations. This similarity also extended to their reactivity to Ca^{2+} (unpublished results).

Only a few electron-dense crystals were associated with negatively stained samples of mineralized liposomes (ammonium molybdate—Fig. 8; phosphotungstic acid—Fig. 9), even though kinetic studies [9, 10] indicated that maximal intraliposomal calcium phosphate formation should have taken place by the time this material was processed for electron microscopy (+5 hours). In contrast, unstained mineralized liposomes showed numerous aggregates of crystals (Figs. 10, 11). The location of this mineral relative to the bilayers was impossible to determine because of the poor definition of the unstained lipid material. The paucity of mineral in the negatively stained samples probably resulted from the preferential dissolution of many crystals during the staining procedure, accelerated by the high affinity of both stains for phosphate [22, 23].

No mineral was identified in thin sections of mineralized preparations fixed with osmium tetroxide alone. However, pretreatment of the samples with tannic acid preserved the mineral deposits while revealing the highly ordered, densely staining lamellar structure of the liposomes (Figs. 12, 13). Small clusters of electron-dense crystallites were present within a substantial number of the vesicles. These mineral inclusions were localized consistently near to, or in contact with, the phospholipid bilayers. More often than not, the crystallites proliferated outwards from their points of contact with the bilayer surface into the aqueous interior of the liposomes (Figs. 13–18). This mineral was quite distinct from the amorphous, electron-dense precipitates of nonspecific tannic acid reaction product present in some samples (Fig. 13). Extraliposomal crystal aggregates, clustered around the outer surfaces of the phospholipid vesicles (Fig. 15), were present following incubation in metastable calcium phosphate solutions [10].

Fig. 1. 7PC:2DCP:1chol liposomes negatively stained with 2% potassium phosphotungstate. Bar = 0.1 μm.

Fig. 2. At higher magnification the lamellar configuration of negatively stained liposomes is evident (2% ammonium molybdate) Bar = 0.1 μm.

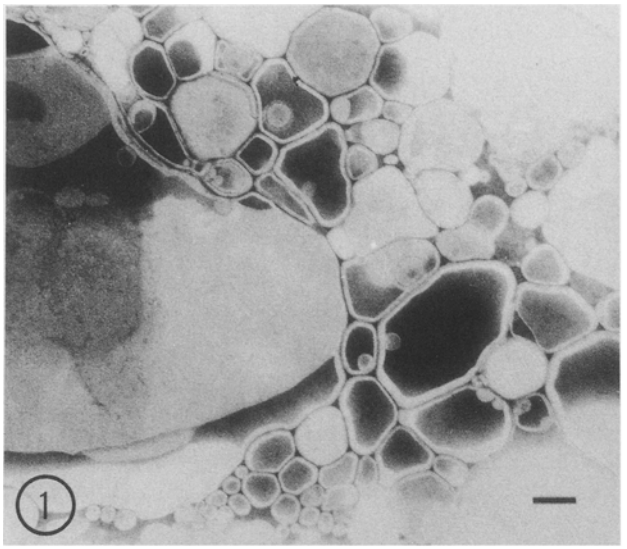
Fig. 3. Thin sections of liposomes fixed with 1% OsO_4. Unstained. Bar = 1.0 μm.

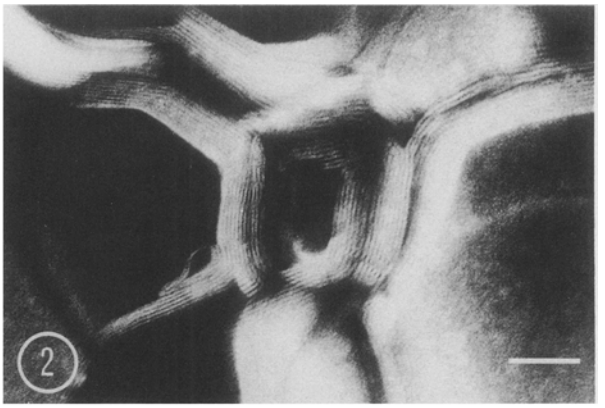
Fig. 4. Numerous lamellae are apparent in osmium-fixed liposomes at higher magnification. Unstained. Bar = 0.5 μm.

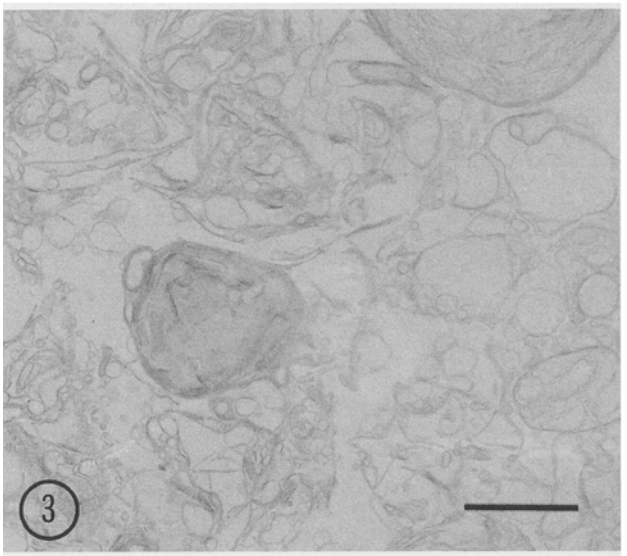
Fig. 5. Freeze-fracture TEM image of smooth-surfaced multilamellar liposomes. Bar = 1.0 μm.

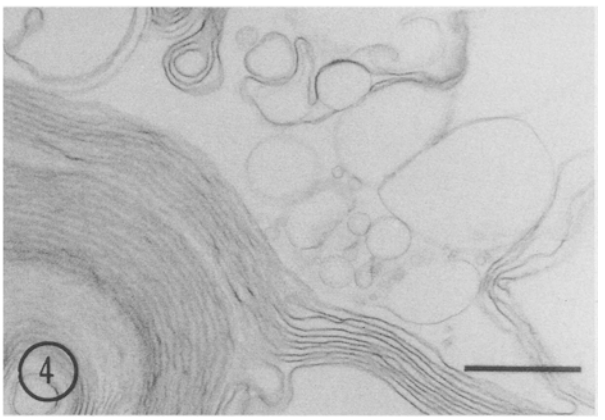
Fig. 6. In freeze-fractured samples, smaller vesicles trapped within the aqueous compartments of larger liposomes are noted. Bar = 1.0 μm.

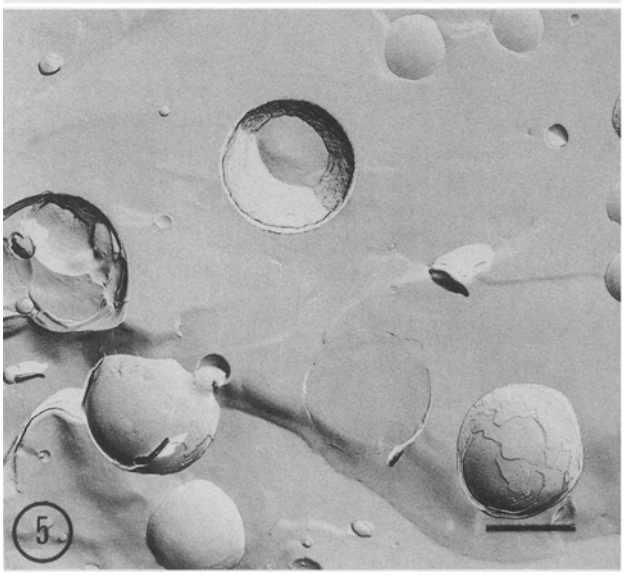
Fig. 7. Cross-fractured liposomes show numerous concentric lipid bilayers. Bar = 0.5 μm.

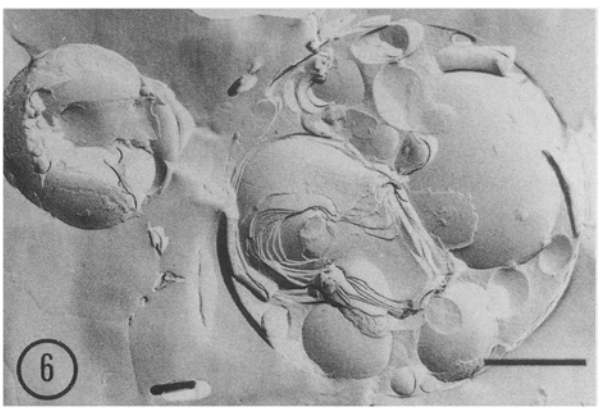

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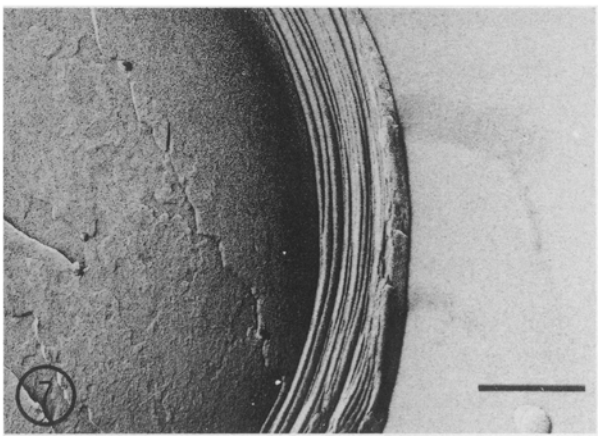

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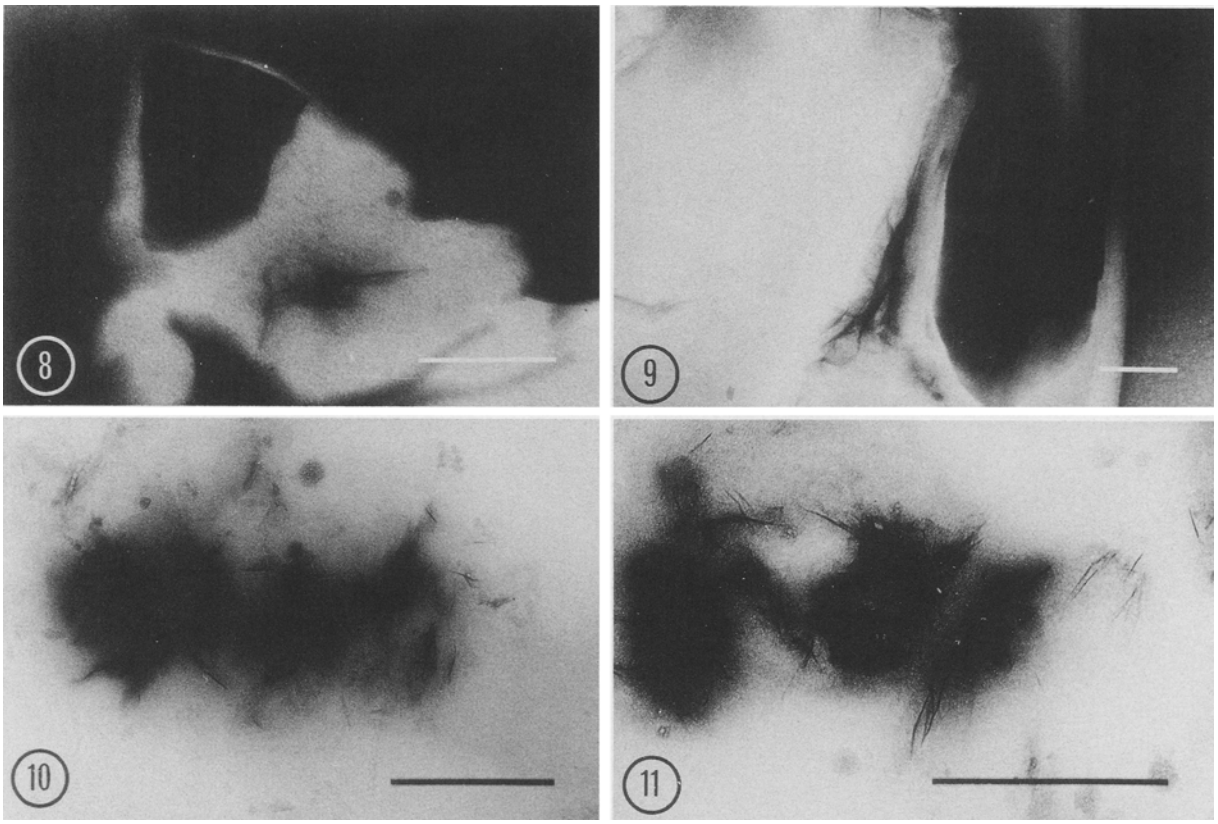


Fig. 8. After ionophore-mediated Ca^{2+} loading, a few indistinct crystals are present in negatively stained liposomes (2% ammonium molybdate). Bar = 0.5 μm.

Fig. 9. Following negative staining (2% potassium phosphotungstate), a few crystal profiles are associated with the X-537A primed liposomes after Ca^{2+} loading. Bar = 0.1 μm.

Fig. 10. Unstained samples of X-537A primed liposomes after Ca^{2+} loading. Numerous crystals of calcium phosphate are evident. Bar = 0.5 μm.

Fig. 11. Mineral deposits are present in unstained samples of Ca^{2+}-loaded PO_4-encapsulated liposomes. The morphology of the lipid vesicles cannot be discerned. Bar = 0.5 μm.

The crystals identified within sectioned material measured approximately 3–6 nm in width, with estimated lengths of 50–350 nm. The apatitic nature of the mineral was confirmed by selected area electron diffraction of unstained sections (data not shown).

In freeze-fractured samples of mineralized preparations, small linear profiles were seen on the ex-

Fig. 12. After tannic acid-OsO_4 treatment, the hydrophilic polar zones of the lipid membranes exhibit intense staining. Bar = 0.05 μm.

Fig. 13. A number of crystalline deposits (*arrowheads*) are present within PO_4-containing liposomes after ionophore-mediated calcium loading. These deposits are quite distinct from the amorphous nonspecific precipitates of tannic acid reaction product (*). Bar = 0.5 μm.

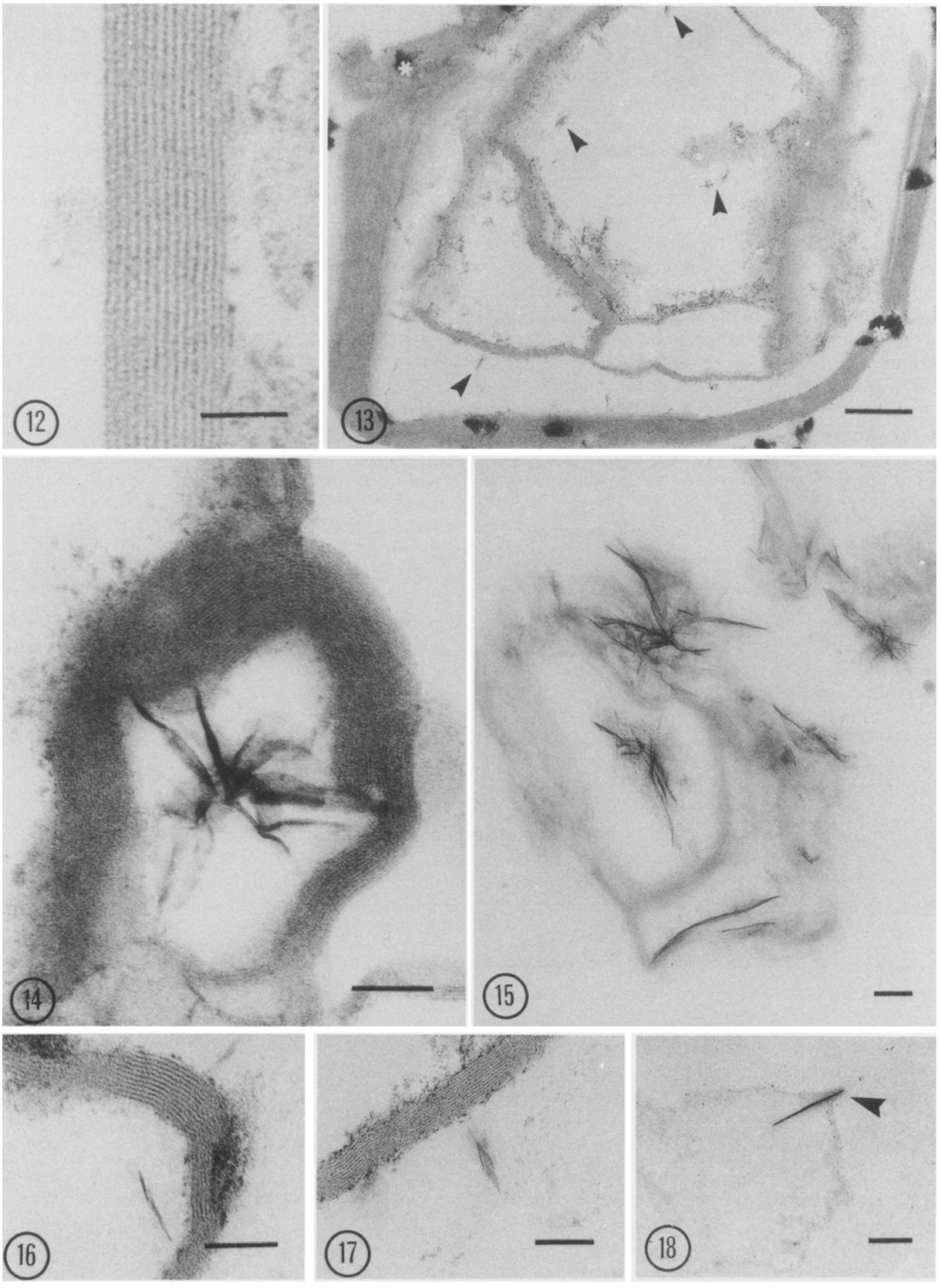
Fig. 14. A PO_4-encapsulated "Liposor" liposome after X-537A-mediated calcium loading. Numerous electron-dense crystals of calcium phosphate are seen within the phospholipid vesicle (tannic acid-OsO_4. lead citrate staining). Bar = 0.1 μm.

Fig. 15. Following incubation in a metastable calcium phosphate solution, extraliposomal crystal aggregates are present (tannic acid-OsO_4). Bar = 0.1 μm.

Fig. 16. Intraliposomal electron-dense mineral deposit close to the inner lipid bilayers of a X-537A primed, Ca^{2+}-loaded liposome (tannic acid-OsO_4. lead citrate staining). Bar = 0.16 μm.

Fig. 17. Intraliposomal precipitate of calcium phosphate adjacent to the internal lamella of a lipid vesicle (tannic acid-OsO_4). Bar = 0.1 μm.

Fig. 18. An electron-dense mineral deposit in close contact with the inner lipid bilayer of a liposome after ionophore-mediated Ca^{2+} loading (naphthaloylhydroxylamine-OsO_4). Bar = 0.05 μm.


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posed surfaces of some liposomes (Figs. 19, 20). Clusters of these inclusions adhered to the lamellae of other liposomes (Figs. 22, 23). When the liposomes were incubated in metastable solutions of calcium phosphate, some phospholipid vesicles were obscured by aggregates of these linear structures (Fig. 24). No such linear profiles were observed in control preparation of Ca^{2+}-exposed liposomes that contained either phosphate or ionophore, but not both. Furthermore, the inclusions were quite distinct from the shallow, irregular cleavage defects which sometimes occurred during the fracturing process (Fig. 21) [25]. There was no evidence of fusion or other detectable changes, such as pits or particles, in the bilayer configuration of the liposomes under any of the reaction conditions. These data suggest that the linear profiles were a direct result of *de novo* apatite formation within the liposomes. The presence of similar profiles in replicas prepared from liposome-free bulk calcium phosphate suspensions (Fig. 25) lends support to this thesis.

Discussion

The preparation of lipid-mineral systems for electron microscopy presents a number of specific technical problems. While negative staining provides detailed evidence of the spheroidal, bilayer structure of the liposomes, the changes in salt concentration, pH, and water content caused by adding the stain and then drying are known to effect the preferential dissolution of the mineral phase [22, 23]. In contrast, electron microscopy of unstained liposomes revealed calcium phosphate deposits in the form of electron-dense crystals, but liposome definition is very poor (Figs. 10, 11). Since conventional cross-linking fixatives (glutaraldehyde and osmium tetroxide) do not readily form insoluble complexes with glycerophosphatides lacking primary amine groups [26] such as those used in this study, an extended incubation time (48 hours) is required to produce adequate fixation of the lipid components. However, such prolonged exposure to aqueous media causes varying degrees of dissolution of inorganic calcium phosphate deposits and translocation of unbound calcium ions [27]. In contrast, the short tannic acid fixation schedule (0.5 hours) ensures rapid fixation of the liposomes with minimal exposure of the intraliposomal precipitate to the aqueous media. It has been shown that tannic acid-OsO_4 treatment significantly reduces the translocation of phosphatidylcholine during tissue processing for electron microscopy [19, 28, 29]. It was found here that by forming tannic acid complexes with liposomal PC prior to osmium fixation, excellent liposome preservation could be achieved under conditions that minimized mineral dissolution and preserved the mineral phase as evinced by clusters of apatitelike crystals within the aqueous interiors of the liposomes (Figs. 14–17). The limited amount of extraliposomal precipitation associated with vesicles from samples incubated in metastable calcium phosphate solutions (Fig. 15) suggests that mineral formed on the outside of the liposomes is still vulnerable to some dissolution during tannic acid-osmium treatment despite the greatly reduced processing times. (Similar results were obtained by preincubating the mineralizing liposomes with N,N-naphthaloylhydroxylamine, sodium salt (0.1 mM) prior to fixation with osmium (Fig. 18). N,N-naphthaloylhydroxylamine is a highly selective and sensitive reagent which has been used for the cytochemical localization of calcium in artherosclerotic calcification [30].)

The freeze etching technique allowed for the examination of the lipid-mineral dispersions without

Fig. 19. Freeze-fracture TEM image of a mineralized preparation. Note the distinct linear structure (*arrowhead*) on the lamellar surface of the liposome. Bar = 0.05 μm.

Fig. 20. A linear inclusion (*arrowhead*) on the exposed surface of a freeze-fractured, X-537A-primed, liposome following Ca^{2+} loading. Bar = 0.1 μm.

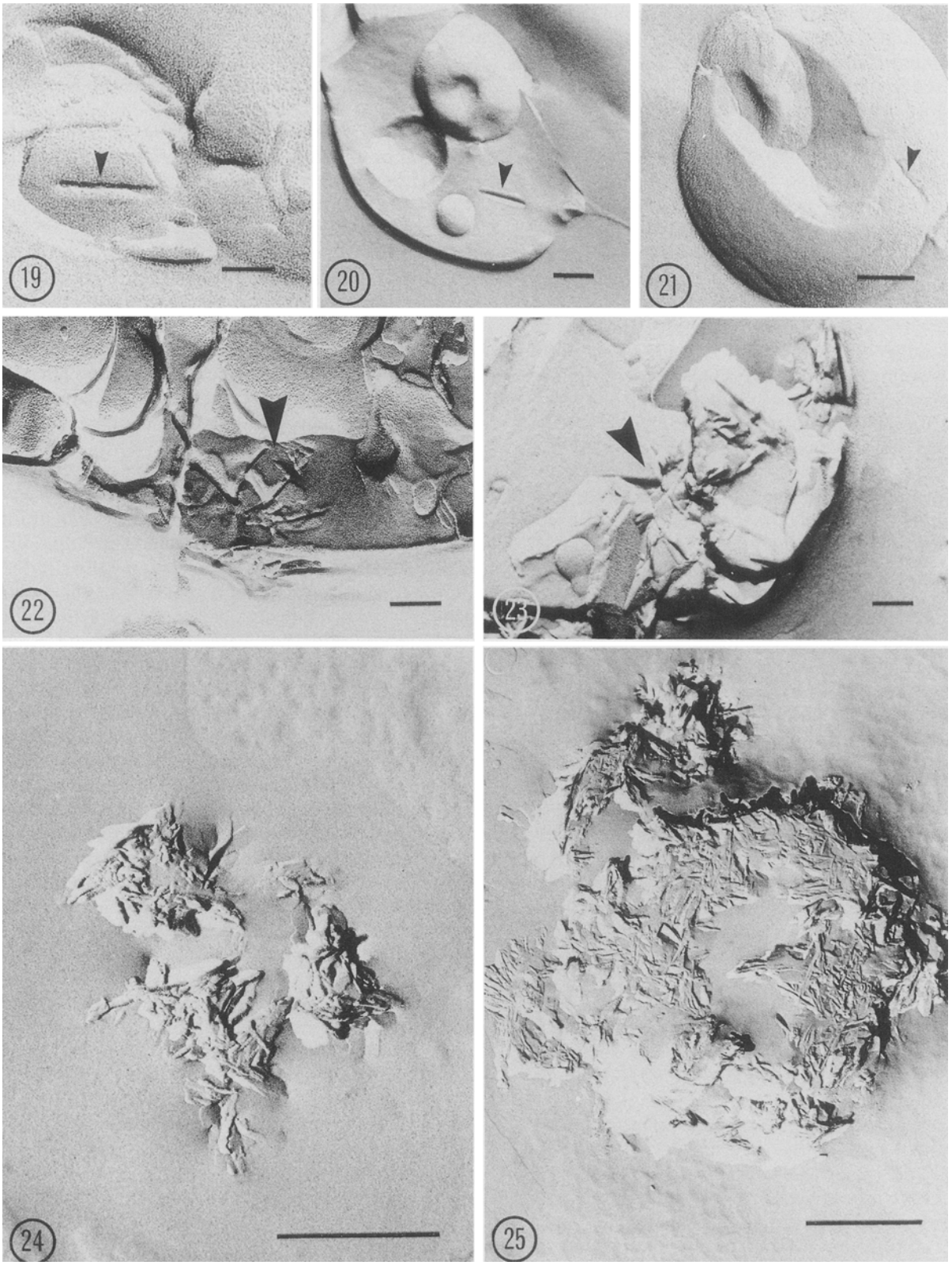
Fig. 21. Freeze-fracture micrograph of PO_4-encapsulated liposome after ionophore-mediated Ca^{2+} loading. A small cleavage defect (*arrowhead*) is present in the exposed bilayer. Bar = 0.1 μm.

Fig. 22. A cluster of linear structures (*arrowhead*) associated with a cross-fractured liposome after intraliposomal calcium phosphate precipitation. Bar = 0.1 μm.

Fig. 23. After ionophore-mediated Ca^{2+} loading, a clump of linear profiles (*arrowhead*) is observed on the exposed lamellar surface of the liposome. Bar = 0.1 μm.

Fig. 24. Freeze-fracture micrograph of liposomes after incubation in a metastable solution of calcium phosphate. The vesicles are obscured by dense aggregates of linear structures. Bar = 0.5 μm.

Fig. 25. Replica of liposome-free calcium phosphate solution. Bar = 0.5 μm.


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recourse to aqueous fixatives. Rapid cooling of the sample to liquid nitrogen temperature was the only pretreatment required. The principal evidence for the presence of mineral was the distinctive linear profiles attached to the otherwise smooth-surfaced lamellae of freeze-fractured liposomes (Figs. 19–20). Their absence from control samples negates the interpretation [31] that these inclusions are the result of separate, specific interactions between adjacent lipid lamellae and Ca^{2+}, phosphate, or ionophore. Furthermore, there was no evidence of these structures arising from the initiation of liposome-liposome fusion events. Rather, it is suggested that these inclusions represent apatite crystals adhering to the lipid bilayers. This argument is supported by the appearance of similar linear profiles in replicas obtained from liposome-free calcium phosphate suspensions (Fig. 25).

Despite the technical difficulties encountered in the preservation of both lipid and mineral phase, electron microscopic analysis of Ca-loaded liposomes clearly revealed the presence of discrete apatite clusters within the lumen and in close proximity to the inner lipid membranes. Ca-loading from metastable calcium phosphate solutions revealed crystal clusters adhering to the exterior liposomal surface. When the incubation conditions precluded intraliposomal precipitation, crystalline apatite phases were not observed.

The above study amplifies and extends the conclusions drawn from earlier kinetic and chemical work on liposome mineralization [9, 10]. Of particular interest was the consistent localization of apatite crystals within the liposomes and the juxtaposition of this mineral to the lipid membrane (Figs. 14–18, 19, 20, 22–24). The apposition of the calcium phosphate deposits to the lamellae may partly explain the increased membrane leakiness of liposomes subjected to intraliposomal precipitation [9]. The formation of crystals in close physical relation to the bilayer would impair the integrity of the membranes and lead to the rupture of the liposomes. (A possible example of incipient rupture is seen in Figure 18.) The mineral thus exposed to the external solution phase would facilitate extraliposomal calcification in metastable calcium phosphate solutions [10]. This would account for the association of apparently external crystal deposits with the liposome structures (Figs. 15, 24). Experiments with a model membrane system of phosphatidylserine vesicles have indicated that while the *de novo* precipitation of amorphous calcium phosphate on the vesicle surface was a prerequisite for vesicle fusion, the transformation of this salt to an apatitelike phase was accompanied by a dramatic increase in membrane leakiness [32]. Also, the deleterious effects of monosodium urate, calcium pyrophosphate dihydrate, silica, and hydroxapatite crystals on cell membranes have been reported in pathological conditions [13–16]. Membrane destablization by these crystalline inorganic salts was induced through localized interactions between the mineral and lipid components in the cell membrane [13, 15, 16]. Direct contact between the crystals and the membrane was an important feature of this process.

There are distinct parallels in the lipid-mineral organization seen in this study and that reported for matrix vesicles [1, 2]. The intraliposomal precipitate formation and subsequent disruption of membrane integrity reinforces the suggestion that the deposition of bioapatite crystals within matrix vesicles initiates the propagation of mineral deposits in the interstitial matrix of calcifying growth plate cartilage.

In conclusion, the complementary use of negative staining, positive staining, and freeze-fracture techniques for studying lipid-mineral interactions arising from calcium phosphate precipitation in the sequestered microenvironment of multilamellar liposomes has provided an experimental vehicle for the *in vitro* study of possible membrane-mediated mineralization events *in vivo*. While the relatively simple composition and multilamellar structure of liposomes contrasts with the chemically complex unit membrane enclosing matrix vesicles, certain aspects of mineral interaction with these membranes can be examined using the liposome model. For example, TEM examination of liposomes may be especially useful for investigating the role of native membrane constituents in establishing and controlling the growth and spatial organization of crystals in and around matrix vesicles during cartilage calcification. The results of a recent study with phosphatidate-containing liposomes have suggested that potential growth sites on crystal surfaces are blocked by the adsorption of membrane-bound monoester phosphate anions to these loci (E. D. Eanes, A. W. Hailer, B. R. Heywood, unpublished data, February 1987).

References

1. Bonucci E (1967) The fine structure of early cartilage calcification. J Ultrastruct Res 20:33–50
2. Anderson HC (1969) Vesicles associated with calcification in matrix of epiphyseal cartilage. J Cell Biol 41:59–72
3. Boskey AL (1981) Current concepts of the physiology and biochemistry of calcification. Clin Orthop Rel Res 157:225–258

4. Anderson HC (1976) Matrix vesicles in cartilage and bone. In: Bourne GE (ed) The biochemistry and physiology of bone, Vol IV. Academic Press, New York, pp 135–155
5. Majeska RJ, Wuthier RE (1975) Studies of matrix vesicles isolated from chicken epiphyseal cartilage. Association of pyrophosphatase and ATPase activities with alkaline phosphatase. Biochim Biophys Acta 391:51–60
6. Fortuna R, Anderson HC, Cartier RP, Sadjera SW (1978) The purification and molecular characterization of alkaline phosphatases from chondrocytes and matrix vesicles of bovine fetal epiphyseal cartilage. Metab Bone Dis Rel Res 1:161–168
7. Wuthier RE, Gore ST (1977) Partition of inorganic ions and phospholipids in isolated cell membrane and matrix vesicle fractions. Evidence for Ca-phospholipid-PO_4 complexes. Calcif Tissue Res 24:163–171
8. Wuthier RE (1976) Lipids of matrix vesicles. Fed Proc 35:117–121
9. Eanes ED, Hailer AW, Costa JL (1984) Calcium phosphate formation in aqueous suspensions of multilamellar liposomes. Calcif Tissue Int 36:421–430
10. Eanes ED, Hailer AW (1985) Liposome-mediated phosphate formation in metastable solutions. Calcif Tissue Int 37:390–394
11. Eanes ED, Hailer AW (1987) Calcium phosphate precipitation in aqueous suspensions of phosphatidylserine-containing liposomes. Calcif Tissue Int 40:43–48
12. Vogel JT (1986) Calcium phosphate solid phase induction by dioleoylphosphatidate liposomes. J Coll Interface Sci 11:152–159
13. Weissmann G, Rita GA (1972) Molecular basis of gouty inflammation: interaction of monosodium urate crystals with lysosomes and liposomes. Nature New Biol 240:167–172
14. Schumacher HR, Fishbein P, Phelps T, Tse R, Krauser W (1978) Comparison of sodium urate and calcium pyrophosphate crystal phagocytosis by polymorphonuclear leukocytes. Arthritis Rheum 18:783–792
15. Kane AB, Stanton RP, Raymond EG, Dobson ME, Knafelc ME, Fraber JL (1980) Dissociation of intracellular lysosomal rupture from cell death caused by silica. J Cell Biol 87:643–657
16. Elferink JGR (1986) Crystal-induced membrane damage: hydroxyapatite crystal-induced hemolysis of erthrocytes. Biochem Med Met Biol 36:25–35
17. Scott K, Tarin D, Sharp JA (1970) Orientation of spherical specimens for ultrathin sectioning in selected planes by embedding agar. J Microsc 91:216–220
18. Spurr AR (1969) A low-viscosity epoxy resin embedding medium for electron microscopy. J Ultrastruct Res 26:31–43
19. Kalina M, Pease DC (1977) The preservation of ultrastructure in saturated phosphatidyl cholines by tannic acid in model systems and type II pneumocytes. J Cell Biol 74:726–741
20. Thorogood PV, Craig-Gray J (1975) Demineralization of bone matrix: observations from electron microscope and electron-probe analysis. Calcif Tissue Res 19:17–26
21. Reynolds ES (1963) The use of lead citrate at high pH as an electron-opaque stain in electron microscopy. J Cell Biol 17:208–210
22. Agostini B, Haselbach W (1971) Effect of some negative stains on the calcium transport of sarcoplasmic reticulum. Histochemie 27:303–309
23. Raeymakers L, Agostini B, Haselbach W (1981) The formation of intravesicular calcium phosphate deposits in microsomes of smooth muscle. A comparison with sarcoplasmic reticulum of skeletal muscle. Histochemistry 70:139–150
24. Bonucci E, Ruerink J (1978) The fine structure of decalcified cartilage and bone: a comparison between decalcification procedures performed before and after embedding. Calcif Tissue Res 25:179–190
25. Zasadzinski JN (1986) Transmission electron microscopy observations of sonication-induced changes in liposome structure. Biophys J 49:119–130
26. Geyer G (1977) Lipid fixation. Acta Histochem [Suppl] (Jena) S209–222
27. Landis WJ (1979) A study of the maintenance of calcium and phosphorous in anhydrous preparations of mineralized tissue for electron optical examination. IN: Lechene P, Warner RR (eds) Microbeam analysis in biology. Academic Press, New York, pp 635–651
28. Saffitz JE, Gross RW, Williamson JR, Sobel BE (1981) Autoradiography of phosphatidylcholine. J Histochem Cytochem 29:371–378
29. Krueger CM, Nuefield EJ, Saffitz JE (1985) Preservation of arachidonyl phospholipids during tissue processing for electron microscopic autoradiography. J Histochem Cytochem 33:799–802
30. Zechmeister A (1979) A new selective ultrahistochemical method for the demonstration of calcium using N,N-napthaloylhydroxylamine. Histochemistry 61:223–232
31. Verkleij AJ, deKruijff B, van Echteld CJ, Noorden PC, Leunissen-Bijvelt J, deGier J (1981) Lipidic particles. Acta Histochem [Suppl] (Jena) S145–149
32. Fraley R, Wilschut J, Duzgenes N, Smith C, Papahadjopoulos D (1980) Studies on the mechanism of membrane fusion: role of phosphate in promoting calcium ion-induced fusion of phospholipid vesicles. Biochemistry 19:6021–6029

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